

The Infrastructure Breakdown

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We are burning engineering cycles and cloud budgets using multi-billion parameter LLMs for data sorting problems a 10-line Python script could handle.

The absolute hardest part of managing technical pipelines right now is not building AI. It is stopping developers from over-engineering basic features.

Consider a real-world system migration:

A team is tasked with building a triage layer for an enterprise support desk receiving 50,000 tickets a day. The requirement is simple: parse the email text and assign it to one of four queues: Billing, Tech Support, Legal, or Account Access.

Instead of deploying a lightweight, local text classification script that processes payloads in 4 milliseconds for fractions of a rupee, the team provisions a complex LangChain pipeline that calls a heavy, commercial cloud LLM on every single incoming webhook.

Jumping straight to Generative AI for deterministic data sorting is a severe architectural failure. It introduces:

- The Cost Tax: Paying by the token for a model to reason about a simple destination folder, spiking API spend.
- The Latency Penalty: Waiting for a public cloud API to process a prompt payload and stream a response takes seconds, creating an instant bottleneck in the data ingestion layer.
- The Reliability Risk: LLMs are non-deterministic. A system that works perfectly on Monday can get creative on Tuesday and route an urgent billing issue to the legal queue for no discernible reason.

This week, we are stripping away vendor marketing hype to look at Traditional Machine Learning versus Generative AI from a systems architecture perspective.

We will break down when to choose math over models, how to spot over-engineering traps early, and a simple 2x2 decision framework to keep production architectures lean.

If you are trying to cut through the AI hype and build cost-efficient production systems, follow along this week.

CTA: Technical leaders, what is the most extreme case of slapping an LLM on a problem that you have seen recently where simple math or traditional code would have sufficed?